

With a specialized master's degree in Image and 3D Technologies, I am passionate about software development, algorithms, and software architecture. Drawing on my experience in 3D vision and robotics, I enjoy designing robust and scalable software. I am looking to expand these skills into other areas of application.



EDUCATION

MASTER'S DEGREE // COMPUTER SCIENCE [2022] **Université Claude Bernard - Lyon 1** Lyon, France
COMPUTER VISION & 3D TECHNOLOGIES

- Analysis, Image Processing, and 3D Computer Vision
- Mesh Generation and Algorithmic Geometry
- 3D Rendering and Realistic Image Synthesis
- Geometric Modeling
- Virtual World Modeling
- Animation, Rigged Models, and Physics Engines
- Advanced Programming
- Project Management and Software Engineering
- Optimization Algorithms
- Web Application Design
- Non-Relational Databases
- Distributed Algorithms

BACHELOR'S DEGREE // COMPUTER SCIENCE [2019] **Université Claude Bernard - Lyon 1** Lyon, France

- Algorithms, Programming, and Data Structures
- Algorithms, Programming, and Complexity
- Applications in Mathematics and Computer Science
- Web Design and Programming
- Advanced Databases
- Networking

EXPERIENCES

COMPUTER VISION & ROBOTICS ENGINEER [2022 - now] **Innodura TB** Lyon, France

- Development of advanced robotics software solutions
 - Robotics applications : bin-picking, automatic hanging, laser stripping, polishing, painting
- Development & integration of **3D vision & AI** algorithms
- Definition & implementation of robotics software architectures
 - Backend architecture (**ROS 2**), frontend web interface (**React.js**, **Three.js**), backend-frontend communication via **WebSocket**
- Robotics : ABB / STÄUBLI
- 3D sensors : ZIVID / SICK / VISIONERF / KEYENCE

INTERNSHIP R&D - ROBOTICS [February 2022 - August 2022] **Innodura TB** Lyon, France

- Integration of path-planning algorithms for mining robot applications and industrial bin-picking applications.
 - Robot Operating System (**ROS**) / **Movelit** / Path Planning / Point Cloud Library (**PCL**) / Octomap / **Voxels**

INTERNSHIP - SOFTWARE DEVELOPMENT [June 2019 - August 2019] **Senzu SAS** Lyon, France

- Development of software for reading & encoding NFC chips.
 - Knowledge of **NFC technology**
 - Software development using **C#** / Windows Presentation Foundation (**WPF**) / Model-View-ViewModel (**MVVM**)

PROJECTS

CITY REPRESENTATION IN UNITY **C++ // Unity // CityGML**

- Modeling the city of Lyon in Unity using CityGML data.
 - Détails : <https://github.com/VCityTeam/DA-POM-VilleUnity>

GEOMETRIC MODELING SOFTWARE **C++ // Qt**

- Software development for modeling Bézier surfaces, surfaces of revolution, and mesh deformation, using QT & C++.

BACKGROUND REMOVAL & REGION GROWING **Python // OpenCV**

- Using OpenCV to remove the background from a video (silhouette extraction) and implementing a region-growing algorithm to segment an image.

CERTIFICATIONS

BECOME A C++ DEVELOPER
LinkedIn Learning

DISCOVER RUST
LinkedIn Learning

QML FOR BEGINNERS
Qt Academy

INTRODUCTION TO QT QUICK 3D
Qt Academy

HOBBIES



CINEMA



VIDEO GAMES



INVESTMENTS



SPACE



VOLLEYBALL

CONTACT



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SKILLS

DOMAINS

Software Engineering
Software Architecture
Algorithmics
Data Structures
Computer Graphics
Computer Vision
Robotics

PROGRAMMING

C / C++
Rust
C#
Java
JavaScript
Python

TECHNOLOGIES

Git
Docker
Qt / QML
Windows Presentation Foundation (WPF)
OpenCV
Open3D
Point Cloud Library (PCL)
Unity
OpenGL
Robot Operating System (ROS) / Movelit
MVTec HALCON
React.js
Three.js

SOFT SKILLS

Autonomy
Creativity
Curiosity
Rigor
Tenacity

LANGUAGES

French (native)
English (professional, TOEIC 965)